

Statistical Power Analysis for the Behavioral Sciences (2nd ed.). by Jacob Cohen
Review by: Peter A. Lachenbruch
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Dempster, A. P. (1988), "Employment Discrimination and Statistical Science" (with discussion), *Statistical Science*, 3, 141–195.

Statistical Power Analysis for the Behavioral Sciences (2nd ed.).

Jacob Cohen. Hillsdale, NJ: Lawrence Erlbaum, 1988. xxi + 567 pp. $55.

This is the second edition of a work that has become a minor classic. It ranks as a required reference for research grant applications in the social sciences, and is largely responsible for concern with sample-size determination in such disciplines. It provided a much-needed reference for those not as numerically proficient as many JASA readers. Not all of the following comments are addressed to those with less-mathematical backgrounds, however.

Cohen begins by discussing the ideas of power analysis, including significance level, reliability of sample results and sample size, effect size, and "types of power analysis" (power as a function of other parameters, sample-size requirements, and detectable effect size). The remaining chapters give a series of tables (including power and needed sample size) for numerous test procedures: the t test, Pearson correlation, difference between correlation coefficients, one-sample test that a proportion is .5, chi-squared tests for goodness of fit and contingency tables, analysis of variance, analysis of covariance, multiple regression, multivariate methods, and tests for differences between proportions.

Wording might have been tighter in some minor areas, but overall the book is clearly written. To the author's credit, references are given thoroughly. Cohen's definitions of "small," "medium," and "large" effect size have proven handy. Even though such definitions are obviously subjective and depend on the context of the problem, some guidelines are good to have; unfortunately, in practice guidelines are often turned into ironclad laws. Cohen relegates all of the formulas to the book's last chapter, with the apologia that "since this is a handbook intended for behavioral scientists, the computational procedures used to determine the power and sample size values of the tables were not given in the previous chapters so as not to interrupt the flow of the exposition of concepts and methods of application" (p. 543). I would prefer that formulas appear in the appropriate chapters.

Other remarks. The t test is standard and taught in elementary courses; the correlation test is based on the t test. Cohen suggests that the failure of the normality assumption "will not seriously affect the validity of significance tests" (p. 75). Kowalski (1972) presented evidence to the contrary, demonstrating serious problems with significance testing for correlations when normality does not hold. Cohen indicates that the Fisher z is used for power and sample size for testing differences in correlation. The test for proportion equal to .5 uses the Harvard tables of the cumulative binomial for $n \leq 50$ and the normal approximation for $n > 50$. The test for differences in proportions uses the arcsin–square root transformation. The tables for the chi-squared tests may be the most useful, since they may be the least familiar.

My primary objection to this 590-page book is that the same result could be achieved more quickly and conveniently by a well-documented set of formulas for a spreadsheet meant to run on a microcomputer, or by a template for the various methods on a floppy disk, accompanied by material from Chapters 1, 11, and 12. Cohen mentions several of the sample-size programs already available on disk in his introduction, and Michael Borenstein and Cohen have written a menu-driven program that performs the computations specified in Cohen's book. This program will allow readers to tailor the sample-size computations precisely to their needs.

PETER A. LACHENBRUCH
University of California, Los Angeles

REFERENCE

Kowalski, C. J. (1972), "On the Effects of Non-Normality on the Distribution of the Sample Product-Moment Correlation Coefficient," *Applied Statistics,* 21, 1–12.

How Many Subjects? Statistical Power Analysis in Research.

Helena Chmura Kraemer and Sue Thiemann. Newbury Park, CA: Sage Publications, 1987. 120 pp. $17.95.

Power analysis and sample-size calculations have received great attention in recent years: By now, assurances of adequate sample size are de rigueur, especially in grant proposals. In his influential book (first published in 1969), Cohen (1988) emphasized the need for power analysis; more important, he made power calculations possible for researchers in a variety of fields. Kraemer and Thiemann's quite different approach [first presented in Kraemer (1985)], which streamlines calculations and makes approximate answers easy to obtain, is a further advance in this direction. They transform each power calculation problem into a problem involving the distribution of the intraclass correlation coefficient.

The book contains 10 chapters. The first two present the fundamental concepts and motivate the practical relevance of what follows. The mathematical framework is given in Chapter 3. Specifically, let (X, Y) have a bivariate normal distribution with correlation ρ and equal variances, and let r be the intraclass correlation coefficient estimated on the basis of a sample of size n. Under these assumptions, the distribution of $u = (r - \rho)/(1 - r\rho)$ depends only on n. In fact, it can be shown that $v^{1/2}u(1 - u^2)^{-1/2}$ has a central t_v distribution, where $v = n - 1$. Consider now the α-level test of $\rho \leq 0$ versus $\rho > 0$, that is, the test that rejects if $r > u_v(\alpha)$, where $u_v(\alpha)$ is the upper-α percentile of the distribution of u. Some algebra shows that if p is the power of this test at an alternative ρ, then $[u_v(\alpha) - u_v(p)]/[1 - u_v(\alpha)u_v(p)] = \Delta$, where $\Delta = \rho$ is, in meta-analytic terminology, a "critical effect size." This last equation is the basis for the construction of Kraemer and Thiemann's master table, which gives the values of v (rounded up to the next integer) that solve the equation for a particular combination of α, p, and Δ.

Except for Chapter 10, in which the authors summarize and offer conclusions, the remaining chapters are devoted to common testing situations: equality of two or more means (Chap. 4), correlation and linear regression (Chaps. 5 and 6), homogeneity of variances (Chap. 7), binomial proportions (Chap. 8), and contingency tables (Chap. 9). The authors bring each problem into the intraclass correlation setting via suitable transformations and then derive the required sample size from the master table. Obviously, the testing situations explicitly covered in this book are mostly very simple. Various widely used methodologies do not appear. For instance, survival analysis is not covered, and the material on multiple regression is limited essentially to testing for the significance of a single predictor. This is not a criticism of the book, only a clarification for prospective readers.

Kraemer and Thiemann's method streamlines and considerably simplifies the calculations for power and sample size. Their book goes beyond "how to" questions; however, the discussion in each chapter is filled with valuable observations and practical suggestions. I found the comparison of several designs in Section 4.4 and the material on setting sample size for contingency tables in Chapter 9 especially useful and enjoyable to read. The authors write in a refreshingly informal and nontechnical style throughout, only occasionally lapsing into vagueness or didacticism ["One should always choose the most powerful statistical test that is valid for the question and the data in hand" (p. 98); "Frequently, what seems intuitively obvious is not only not obvious, but not even true" (p. 99)].

In attempting to inform readers as various as the professional statistician and the individual only distantly acquainted with statistics, the authors obviously cannot do justice to every reader's needs and are likely to be criticized whatever approach they take. For example, in my judgment Kraemer and Thiemann have perhaps simplified power calculations too much. By making power calculations seem so easy, they leave room for misunderstanding and, possibly, for misapplication. I would welcome a more balanced mix of theory and applications. For example, the terse explanation of the derivation of the master table in Chapter 3 is presumably meant for statistically minded readers. Other readers are assured that they may skip this chapter "without compromising the understanding of what follows" (p. 28). But what is there to understand in most of what follows, other than how to apply formulas that allow the use of the master table? Kraemer and Thiemann rarely explain the derivation of such formulas and rarely discuss how adequate the approximations involved in using the master table are. Statistically sophisticated readers would profit from such explanations, whereas the rest would at least understand that there is more here than meets the eye and that they should use this methodology with care!

As the following quibble about Section 6.2 ("Regression Versus Correlation") illustrates, glossing over technical points may go too far. The authors write:

> If, instead of sampling certain fixed X values, the researcher chooses to draw a random sample from a bivariate normal population, then either the regression or the correlation approaches can be applied. Which approach has more power? Neither does of course: both have the same power. (p. 62)

At best, this passage is confusing, since the two models are not inter-

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